

ECE501 - Digital Image Processing

Weekly Report: Hybrid Multi-Frequency Image Illusion

Week 4

Group 10

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1. Overview of the past Week

Week 3 we performed research on quantitative analysis and effectiveness on parameter variation through it. Collected some pair of images for testing.

2. Work Completed During the Week

In week4, following the continuation of week3, our plan was to expand dataset over various other features and perform parameter variation on common ground using the results of quantitative analysis and try same approach with new dataset to setup for end goal of image illusion in following 1-2 weeks.

- We first studied past weeks results from variations in perciously discussed parameters (like kernel size, gaussian blur SD).
- Further we made conclusion from our findings and variations in those paramters.
- Later studied and collected image with similarity so that set of parameter values are most effective in image illusion.

3. Key Observations

- Currently, the results are noticable but with specification like particular feature rich dataset will only be implied with best results.

- The quantitative analysis over with techniques also helps to know the problems and changes in features of image illusion because of change in combination set of parameters.

4. Challenges Faced

The challenges were kind of similar to week 3 because we performed the extension of work from week3 but specifically, the image clarity when observed from point of view of background and lighting contrast was a major challenge through this week too.

5. Work Planned for Next Week

Trying to improve the quality of image illusion for wide range over different characteristics such that it matches both our subjective and objective targets in following week .

6. Conclusion

This week was extension of precious week due to diwali break, but our group tried to pull out manageable time. Moreover, we expanded daataset and made clarity about parameter's effectiveness and dataset that shows good result for that particular value of parameters and why it was so. This gave us good understanding of our class discussion and lecture tutorials through practical implementation in our project like MSE and SSIM.